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Integrating Upstream Supply Chains into Generation Expansion Planning

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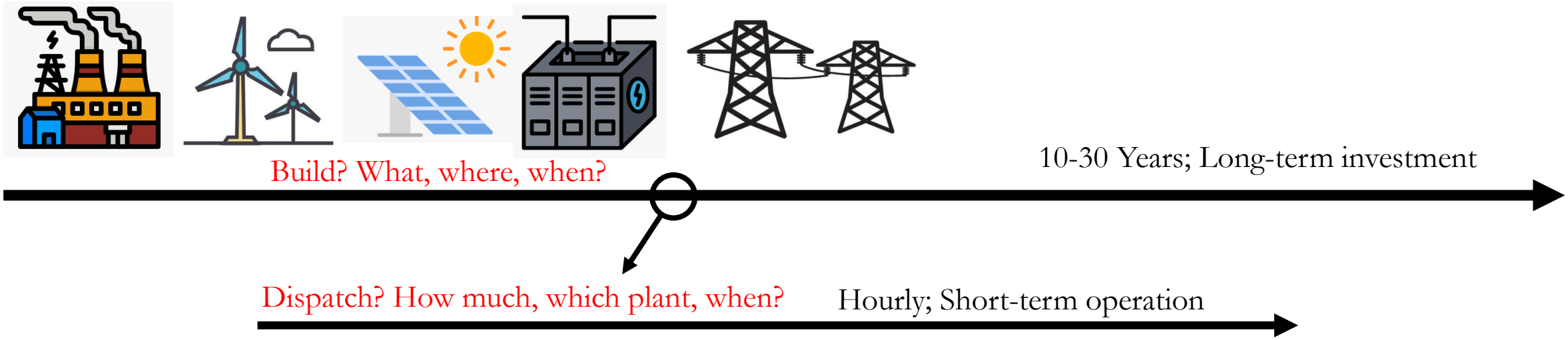
What is Capacity Expansion Planning

- **A long-term planning tool** decides what power plants, storage, and transmission to build, where, and when.
- **Balances trade-offs** among cost, reliability, and policy-driven goals under an uncertain future.
- **Integrates multiple time scales** i.e., short-term operations with long-term investment choices.
- **Informs policy and markets** by guiding regulators, utilities, and system operators on infrastructure and climate targets.



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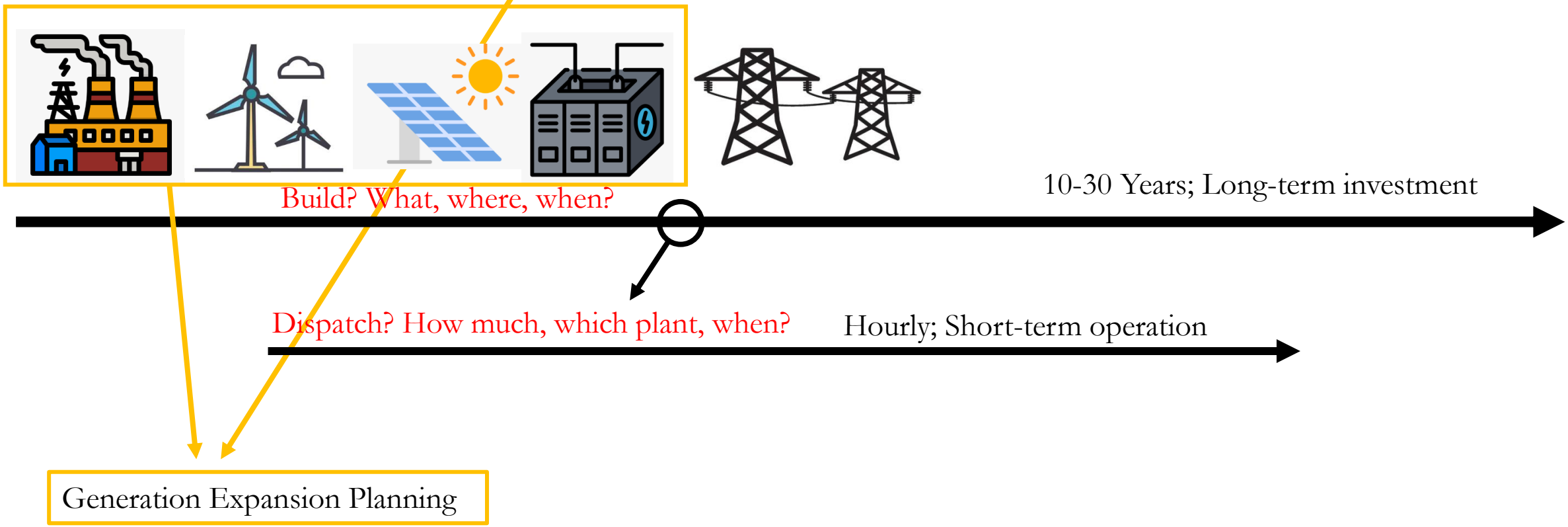
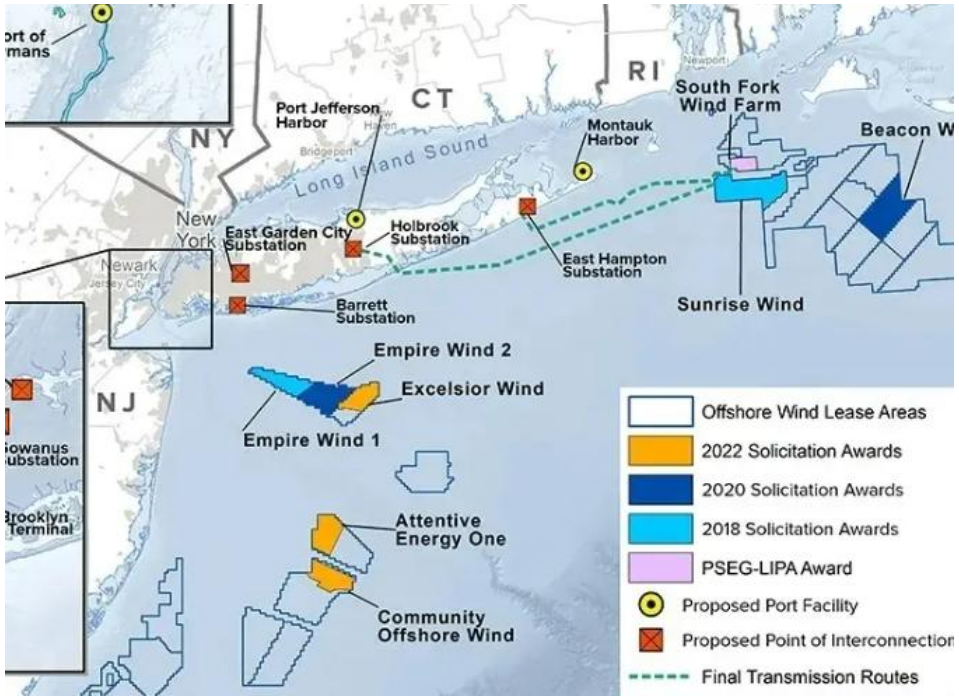


2 Stories

- Story 1 — The Broken Promise (New York Wind)
  - **2021:** New York awarded contracts for massive offshore wind farms (e.g. *Empire Wind 2*), meant to power millions of homes [1].
  - Contracts locked in, market signals clear, developers ready to build.
  - **By 2024:** plans collapsed — inflation, material bottlenecks, and shortages of submarine cables and turbines [2].
  - The contract was torn up: the market promised clean megawatts, but the supply chain couldn't deliver.

- Story 2 — The Price Shock (PJM Capacity Market)
  - **2024:** PJM capacity auctions saw prices spike up to **800%** in some zones [3].
  - Consumers and businesses faced billions in extra costs.
  - "We need new capacity, now!"
  - Projects stalled in queues, with transformers, materials, and financing all in short supply.



[1] https://www.nyserda.ny.gov/All-Programs/Offshore-Wind/Focus-Areas/NY-Offshore-Wind-Projects
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[3] https://www.reuters.com/business/energy/pjm-power-auction-results-yield-sharply-higher-prices-2024-07-31
[4] https://opc-dc.gov/wp-content/uploads/2025/05/PJM-Capacity-Market-Report-FINAL-OPC-Synapse.pdf

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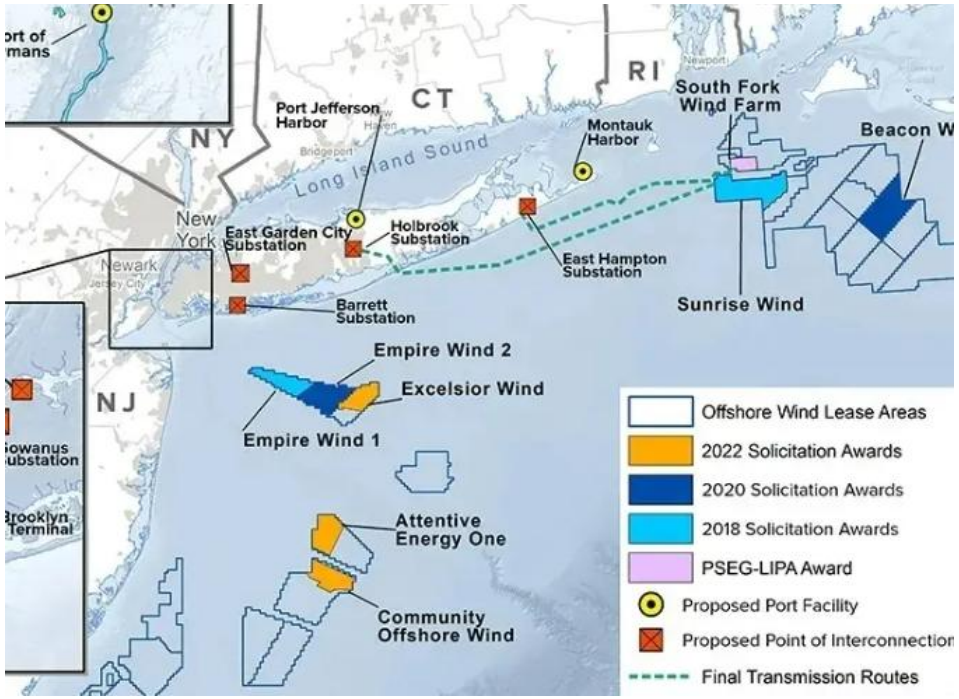
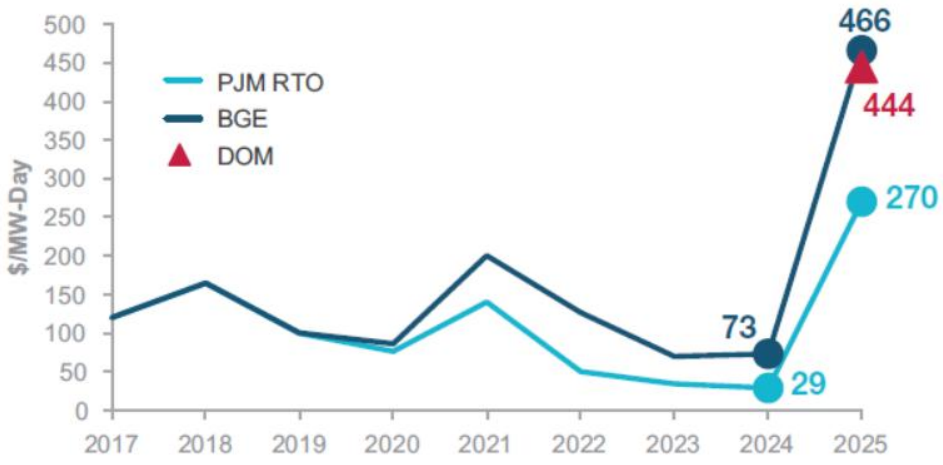
Why? Mismatch between planning and supply chain reality!





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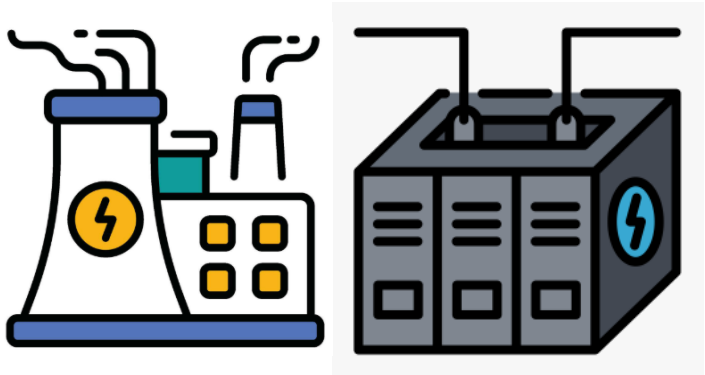
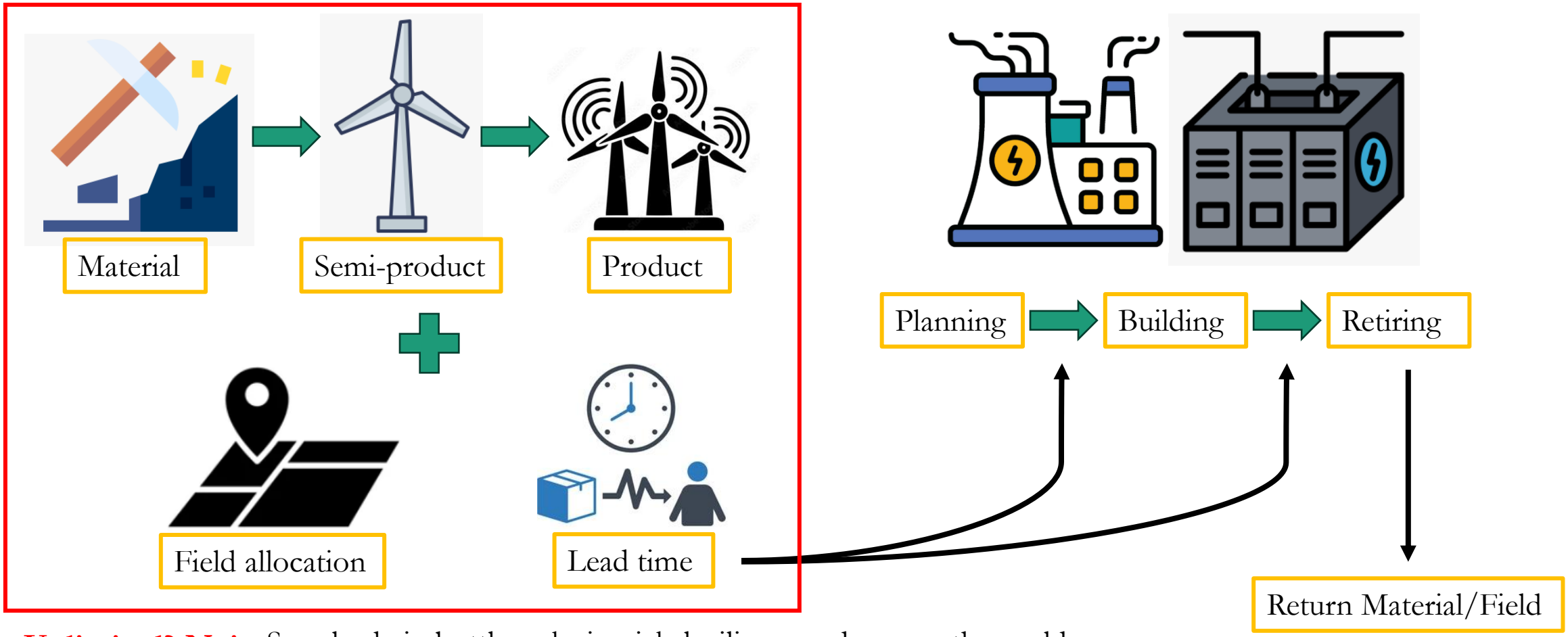
Problem Description

- **Traditional Capacity Expansion:** focuses on deciding when, where, and what size to build, without considering upstream supply chain availability.



Planning → Building → Retiring

Problem Description



Unlimited? No! Supply chain bottlenecks in nickel, silicon, and rare earths could leave the U.S. 730 GW short — 34% of 2050 clean energy goals [5].

[5] Yao, B., Jeong, H., Mehrtash, M., Allan, B., Ockerman, D., Dvorkin, Y. (2025). Understanding supply chain constraints for the US clean energy transition. npj Clean Energy, 1(9). Https://www.nature.com/articles/s44406-025-00009-1

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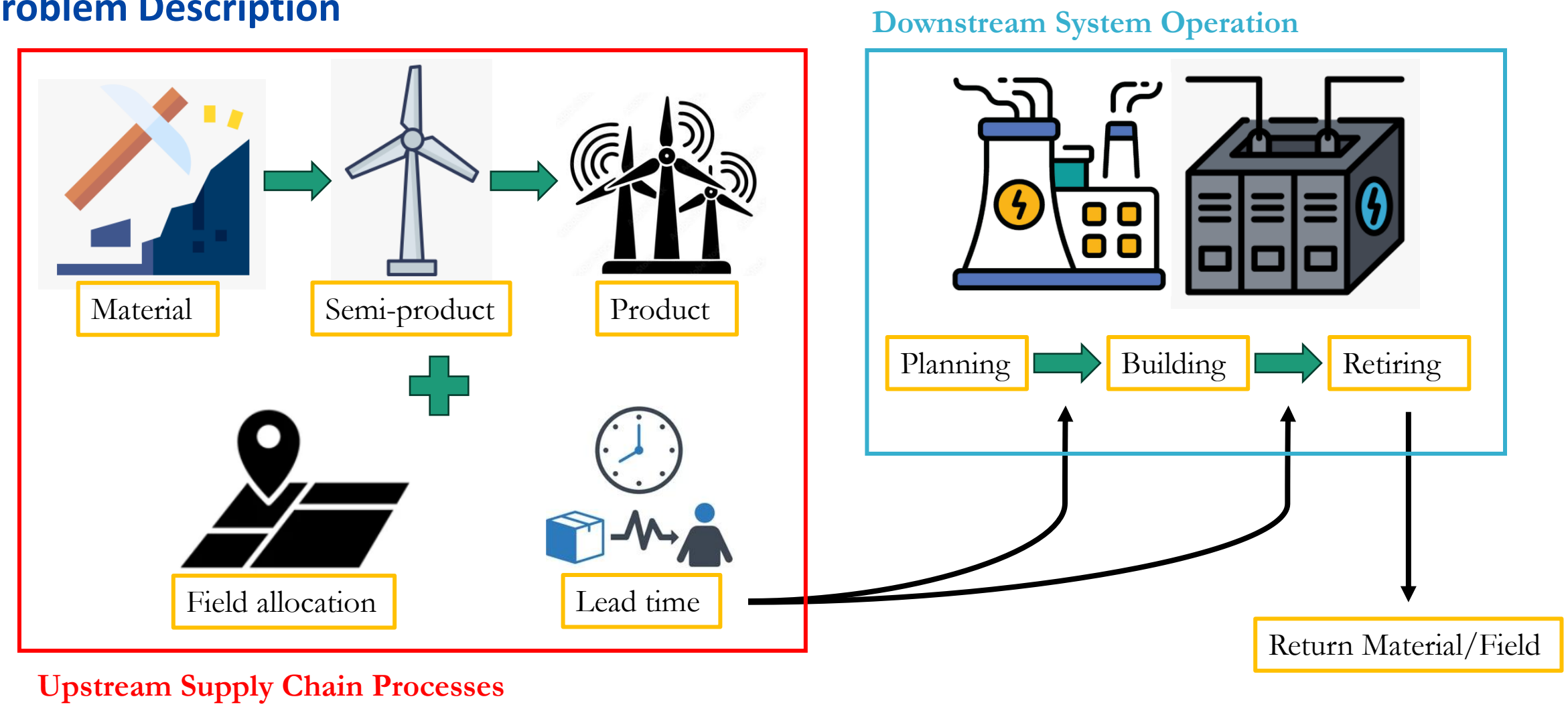
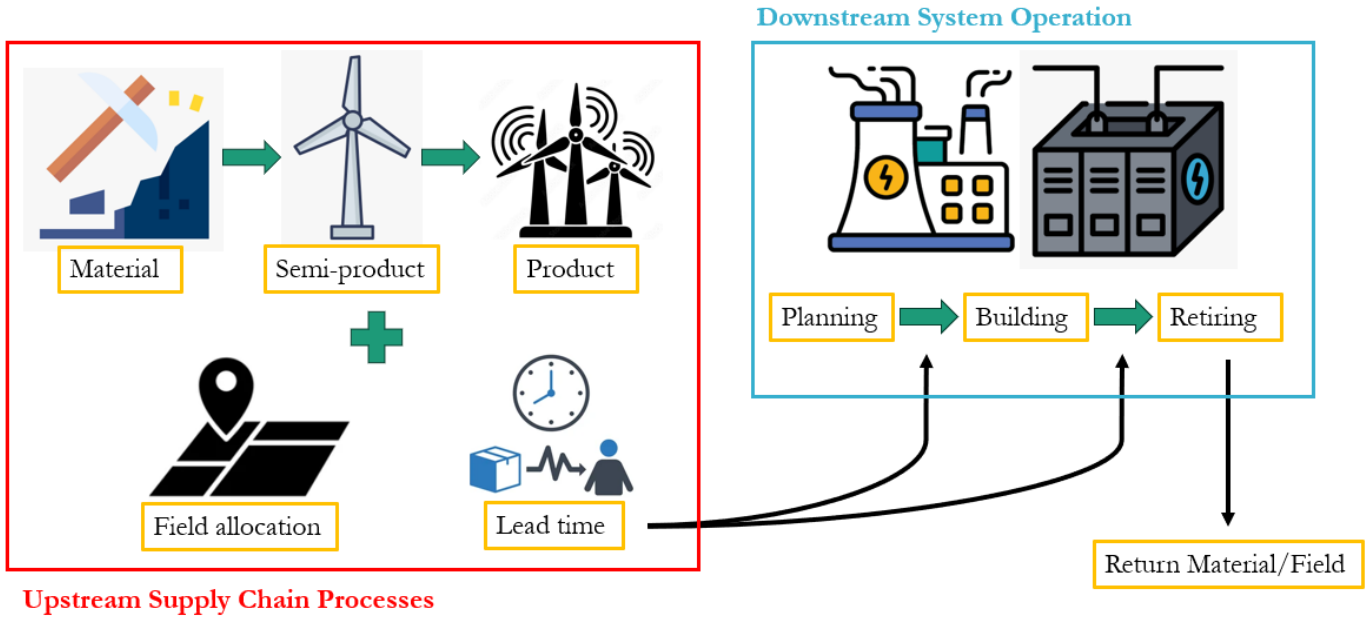


Problem Description

- **Decisions (annual):** where/when/what to build or retire; product mix (e.g., SPV, LBW/OSW, BSS variants).
- **Operations (hourly, rep. days):** dispatch, storage charge/discharge, flows, shedding
- **Upstream layers:** materials → semi-products → products mapped to capacity; stocks & recovery from retirements; field availability updates over time.



$$\min \sum_y C_y = \sum_y \left(C_y^{\text{in}} + C_y^{\text{op}} + C_y^{\text{pe}}\right) \quad (1a)$$

$$C_y^{\text{in}} = \sum_{g \in \bar{\mathcal{G}}} AC_{gy}^{\text{I}} \cdot \overline{P}_g^{\text{G}} \cdot d_{gy} \quad (1b)$$

$$C_y^{\text{op}} = \sum_g C_g^{\text{F}} \cdot \overline{P}_g^{\text{G}} \cdot o_{gy} + \sum_{g \in \mathcal{G}^{\text{th}} \cup \mathcal{G}^{\text{rn}}} \sum_t C_g^{\text{V}} \cdot N_{ty} \cdot \sum_h p_{gthy} + \sum_{g \in \mathcal{G}^{\text{st}}} \sum_t C_g^{\text{V}} \cdot N_{ty} \cdot \sum_h (c_{gthy} + dc_{gthy}) \quad (1c)$$

$$C_y^{\text{pe}} = \sum_t N_{ty} \cdot P^{\text{VOLL}} \cdot \sum_h p_{ithy}^{\text{LS}} + P^{\text{RM}} \cdot p_y^{\text{RM}} + P^{\text{RPS}} \cdot e_{ky}^{\text{RPS}} \quad (1d)$$

$$u_{my} \geq \sum_c v_{cy} \cdot D_{mc}^{\text{CO}}, \quad \forall m, y \quad (2a)$$

$$v_{cy} \geq \sum_p w_{py} \cdot D_{cp}^{\text{PR}}, \quad \forall c, y \quad (2b)$$

$$u_{my} \leq \overline{M}_{my} + \sum_{g \in \mathcal{G}} R_{mg}^{\text{RM}} \cdot \overline{P}_g^{\text{G}} \cdot r_{gy} + s_{my}, \quad \forall m, y \quad (2c)$$

$$s_{my} = s_{m(y-1)} + \overline{M}_{m(y-1)} - u_{m(y-1)} + \sum_{g \in \mathcal{G}} R_{mg}^{\text{RM}} \cdot \overline{P}_g^{\text{G}} \cdot r_{g(y-1)}, \quad \forall m, y \quad (2d)$$

$$\sum_{g \in \mathcal{G}^k} \overline{P}_g^{\text{G}} \cdot d_{gy} \leq \sum_{p \in \mathcal{P}^k} w_{py}, \quad \forall k, y \quad (2e)$$

$$\frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot d_{gy}}{R_k^{\text{CAP}}} \leq f_{iy}^k, \quad \forall i, y, k \quad (2f)$$

$$f_{iy}^k = f_{i(y-1)}^k + \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot r_{gy}}{R_k^{\text{CAP}}} - \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot d_{g(y-1)}}{R_k^{\text{CAP}}}, \quad \forall i, y > 1, k \quad (2g)$$

$$f_{i1}^k = A_i^k + \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot r_{g1}}{R_k^{\text{CAP}}}, \quad \forall i, k \quad (2h)$$

$$\sum_{y'' \leq y - T_g^{\text{LEAD}}} d_{gy''} = \sum_{y' \leq y} b_{gy'}, \quad \forall g \in \bar{\mathcal{G}}, y \quad (2i)$$

$$b_{gy} = 0, \quad \forall g \in \bar{\mathcal{G}}, y \quad (2j)$$

$$\sum_{y'' \leq y - T_g^{\text{LT}}} b_{gy''} = \sum_{y' \leq y} r_{gy'}, \quad \forall g \in \bar{\mathcal{G}}, y \quad (2k)$$

$$r_{gy} = 1 \text{ if } y = T_g^{\text{RT}}, \ 0 \text{ if } y < T_g^{\text{RT}}, \quad \forall g \in \bar{\mathcal{G}}, \forall y \quad (2l)$$

$$\sum_{g \in \mathcal{G}_i \cap (\mathcal{G}^{\text{th}} \cup \mathcal{G}^{\text{rn}})} p_{gthy} + \sum_{g \in \mathcal{G}_i \cap \mathcal{G}^{\text{st}}} (dc_{gthy} - c_{gthy}) - \sum_{l \in \mathcal{LS}_i} q_{lthy} + \sum_{l \in \mathcal{LR}_i} q_{lthy} = L_{ithy} - p_{ithy}^{\text{LS}}, \quad \forall i, t, h, y \quad (3a)$$

$$0 \leq p_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{th}}, y, t, h \quad (3b)$$

$$0 \leq p_{gthy} \leq F_{igthy}^{\text{GEN}} \cdot \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}_i \cap \mathcal{G}^{\text{rn}}, i, y, t, h \quad (3c)$$

$$o_{gy} = o_{g(y-1)} + b_{gy} - r_{gy}, \quad \forall g \in \mathcal{G}, y \geq 1 \quad (3d)$$

$$o_{g0} = 1 \text{ if } g \in \bar{\mathcal{G}}, \ 0 \text{ otherwise}, \quad \forall g \in \mathcal{G} \quad (3e)$$

$$-\overline{P}_l^{\text{L}} \leq q_{lthy} \leq \overline{P}_l^{\text{L}}, \quad \forall l, y, t, h \quad (3f)$$

$$\sum_{g \in \mathcal{G}^k} \overline{P}_g^{\text{G}} \cdot F_{ky}^{\text{ELCC}} \cdot o_{gy} + p_y^{\text{RM}} \geq (1 + R_y^{\text{RM}}) \cdot \overline{L}_y, \quad \forall y, k \quad (3g)$$

$$0 \leq p_{ithy}^{\text{LS}} \leq L_{ithy}, \quad \forall i, y, t, h \quad (3h)$$

$$\sum_{g \in \mathcal{G}^k} \sum_t \sum_h N_{ty} \cdot p_{gthy} + e_{ky}^{\text{RPS}} \geq R_{ky}^{\text{RPS}} \cdot \sum_t \sum_h \sum_i N_{ty} \cdot L_{ithy}, \quad \forall k, y \quad (3i)$$

$$0 \leq c_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{st}}, y, t, h \quad (4a)$$

$$0 \leq dc_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{st}}, y, t, h \quad (4b)$$

$$0 \leq e_{gthy}^{\text{soc}} \leq \overline{E}_g^{\text{ST}}, \quad \forall g \in \mathcal{G}^{\text{st}}, y, t, h \quad (4c)$$

$$e_{gthy}^{\text{soc}} = e_{gt(h-1)y}^{\text{soc}} + \epsilon^{\text{CH}} \cdot c_{gthy} - \frac{dc_{gthy}}{\epsilon^{\text{DC}}}, \quad \forall g \in \mathcal{G}^{\text{st}}, y, t, h \quad (4d)$$

$$e_{gt1y}^{\text{soc}} = e_{gtendy}^{\text{soc}} = 0.5 \cdot \overline{E}_g^{\text{ST}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{st}}, y, t \quad (4e)$$

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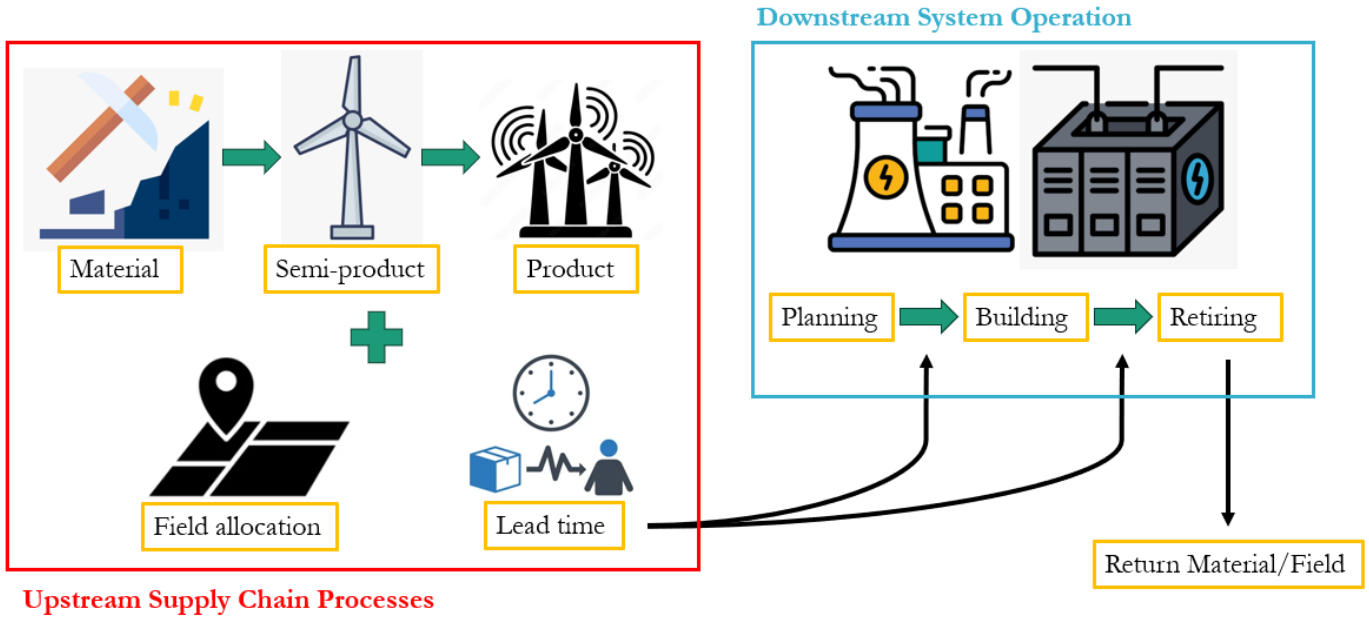

Objective = Investment + Operations + Penalties
Build new capacity, run the system, cover shortfalls

Upstream layer:
1. Tracks materials → components → products
2. Limits by production capacity & lead times
3. Accounts for site availability
4. Recycling from retired units reduces new demand

Downstream layer:
1. **Balance**: generation + imports + storage = demand − shedding
2. **Limits**: generation, line capacity, load shedding bounds
3. **Evolution**: unit builds/retirements over years
4. **Battery:** charge/discharge ≤ rated capacity; SOC within bounds
5. **Reliability**: ELCC ≥ peak + reserves
6. **Compliance**: meet RPS targets (with slack)

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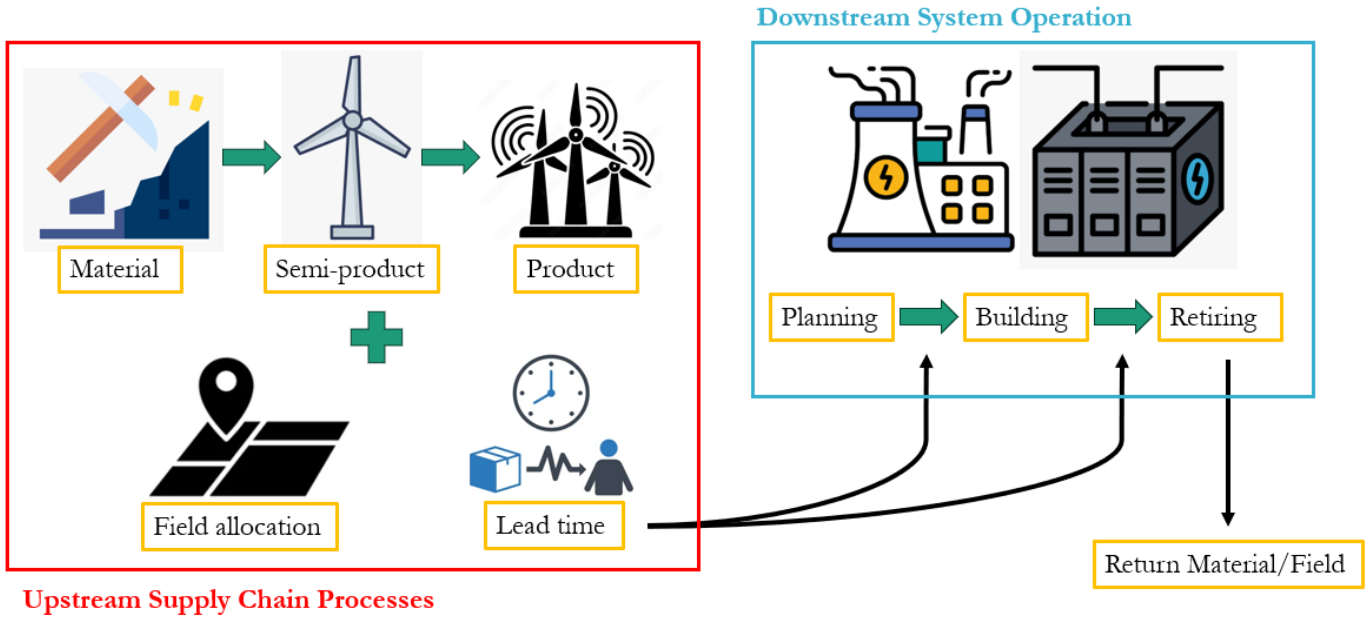
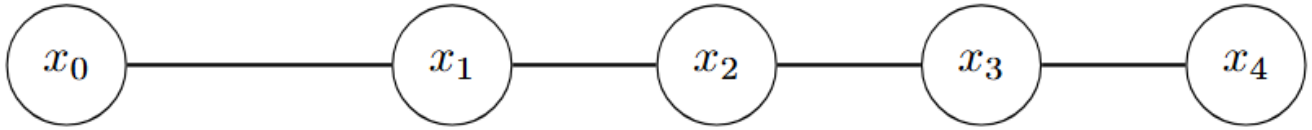


$t = 0$ (year) $\quad t = 1 \quad t = 2 \quad t = 3 \quad t = 4$



$$\min_{(x_1,y_1),\ldots,(x_T,y_T)} \left\{ \sum_{t=1}^{T} f_t(x_t, y_t) \;:\; (x_{t-1}, x_t, y_t) \in X_t,\ \forall t = 1, \ldots, T \right\}$$

Compact form

- x_t: state or linking variable → Cross-year updates: material/field availability, unit operation status.
- y_t: local decision variable → Within-year supply-chain logistics; hourly dispatch and operations.
- $f_t(x_t, y_t)$: stage-wise cost
- X_t: feasible set coupling x_{t-1}, x_t, and y_t

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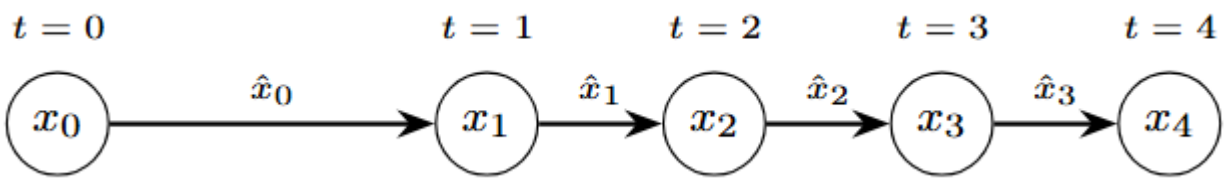
Accelerating Computation with Dual Dynamic Programming



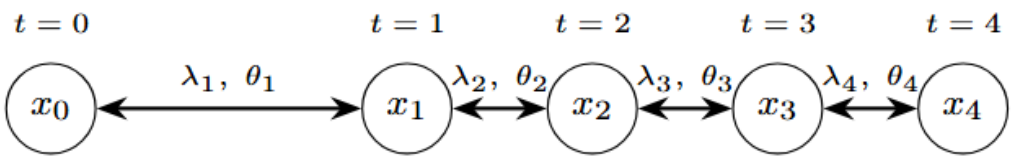
$$\min_{x_1, y_1} \left\{ f_1(x_1, y_1) + \sum_{m \in \mathcal{C}(1)} Q_m(x_1) \ : \ (x_{a(1)}, x_1, y_1) \in X_1 \right\}$$

- Rather than a single extensive-form model over all stages, we solve it recursively by stage. Qm(xt) denotes the cost-to-go, the minimal remaining cost from decision xt.
- In the **forward pass**, given an initial point x_0, we solve:

$$\hat{x}_t = \arg\min_{x_t, y_t} \left\{ f_t(x_t, y_t) + \sum_{m \in \mathcal{C}(t)} \hat{Q}_m(x_t) \ : \ (x_{a(t)}, x_t, y_t) \in X_t \right\}$$

- The solution $\hat{x}_t$ is passed forward and used as input for the next stage.
- At year t, we choose decisions that make today's cost + our best estimate of future cost as small as possible.

Accelerating Computation with Dual Dynamic Programming



- In the **backward pass**, we recursively solve the subproblems starting from the last stage T down to the first.
- At each stage t, solving the subproblem provides:
  - λ_t: dual multipliers from constraints (slope of the cut).
  - θ_t: the optimal objective value (intercept of the cut).
- Solving the subproblem at stage t yields a **cut**, such as a Benders cut based on a linear approximation:

$$\hat{Q}_t(x_{t-1}) \geq \theta_t + \lambda_t^\top x_{t-1}$$

- This cut is valid and tight by strong duality in linear programming.
- As iterations proceed, we collect multiple cuts to approximate the cost-to-go:

$$\hat{Q}_t(x_{t-1}) = \max_{\ell \in \mathcal{L}_t} \left\{ \theta_t^\ell + \lambda_t^{\ell\top} x_{t-1} \right\}$$

- These cuts are passed backward to improve earlier-stage approximations.

- We're "drawing straight cut under the future-cost curve."
- Each new cut touches the curve at one point and never crosses it; the stack of cuts gets closer to the real curve over **iterations**.

Accelerating Computation with Dual Dynamic Programming

- The algorithm alternates between a **forward pass** and a **backward pass**.
- In the forward pass:
  - Given $\hat{x}_{t-1}$, solve the stage t subproblem.
  - Use current approximation $\hat{Q}_{t+1}(x_t)$ to evaluate future cost.
- In the backward pass:
  - Solve the LP at each stage to obtain dual λ_t and cost θ_t.
  - Use these to generate new cuts and update $\hat{Q}_t$.
- Convergence is achieved when the cost estimate stabilizes.

- Make a feasible, stage-by-stage decision using the current stage info and the best cost-to-go estimate so far; pass the state forward.

- From the last stage back, use duals (sensitivities) to add cuts that tighten the cost-to-go; i.e., learn how total cost changes with the state.

Case Study: Maryland

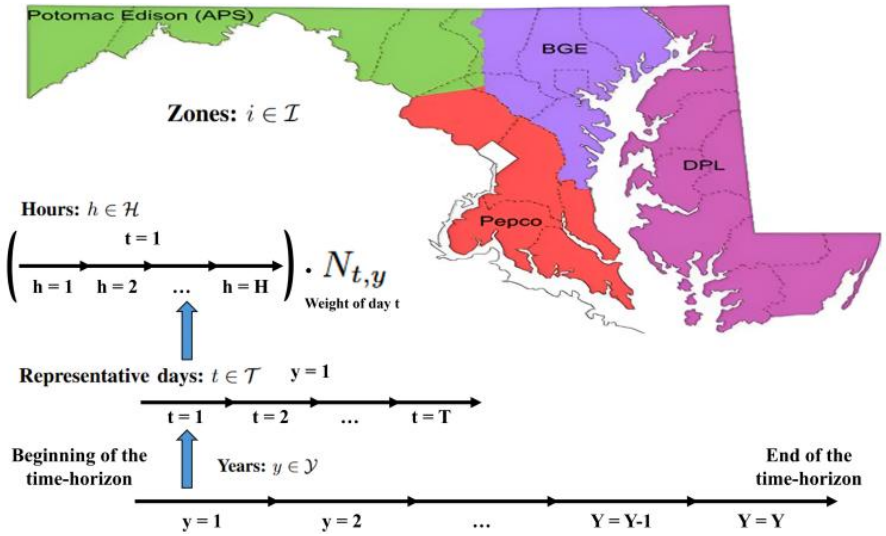


Fig. 1. Maryland spatial and temporal representation

- 4 zones (BGE, APS, DPL, PEPCO) as nodes.
- Investments made annually; operations modeled hourly with 1 representative day per season (via k-means clustering).
- Transmission: 4-node network, PJM Window 3 upgrades assumed, no further expansion.
- Focus: captures supply chain impacts in long-term planning.

Assumptions

- Resources: existing + new (renewables, batteries only) (per MD regs & queue).
- Lifetimes: retire at design end (+1 year if exceeded).
- Materials: 14 critical materials → 12 key products.
- Supply: domestic, imports, stock, 10% recycling.
- Lead times: fixed tech-specific delays.
- Field: land & offshore limits; shared zones; reuse after retirements.

Scenarios:

- **Low**: excludes aggressive electrification & data center growth.
- **High**: includes both electrification & data center growth.
- Sub-Scenarios:
  - **w/o SC** (relaxed): unlimited materials, no lead times, expanded land/offshore areas.
  - **lim. SC** (constrained): materials limited to domestic + allied sources, with lead times & land limits reflecting geopolitical/trade risks

Results

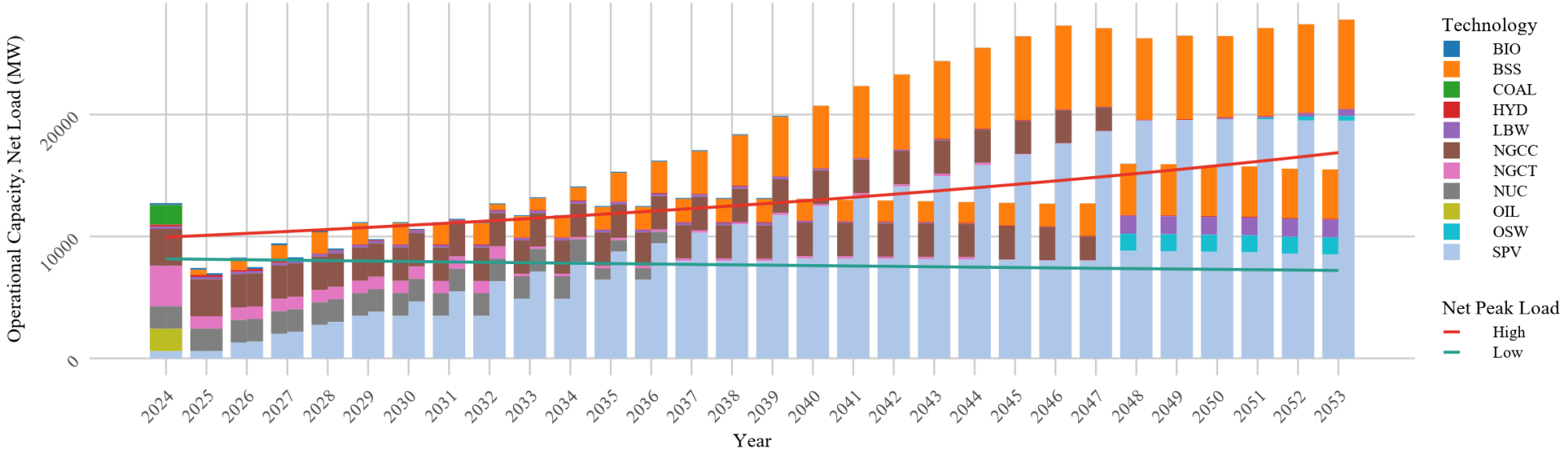


Fig. 2. Operational capacity time for baseline Low and High scenarios. Stacked bars: tech-specific capacity (Low baseline = left); Lines: net peak load (system demand)

- Low baseline (2053): 4.1 GW BSS, 1.5 GW LBW, 1.4 GW OSW, 8.5 GW SPV → more diversified mix.
- High baseline (2053): 7.3 GW BSS, 0.5 GW LBW, 0.4 GW OSW, 19.5 GW SPV → solar-dominant growth.

Results

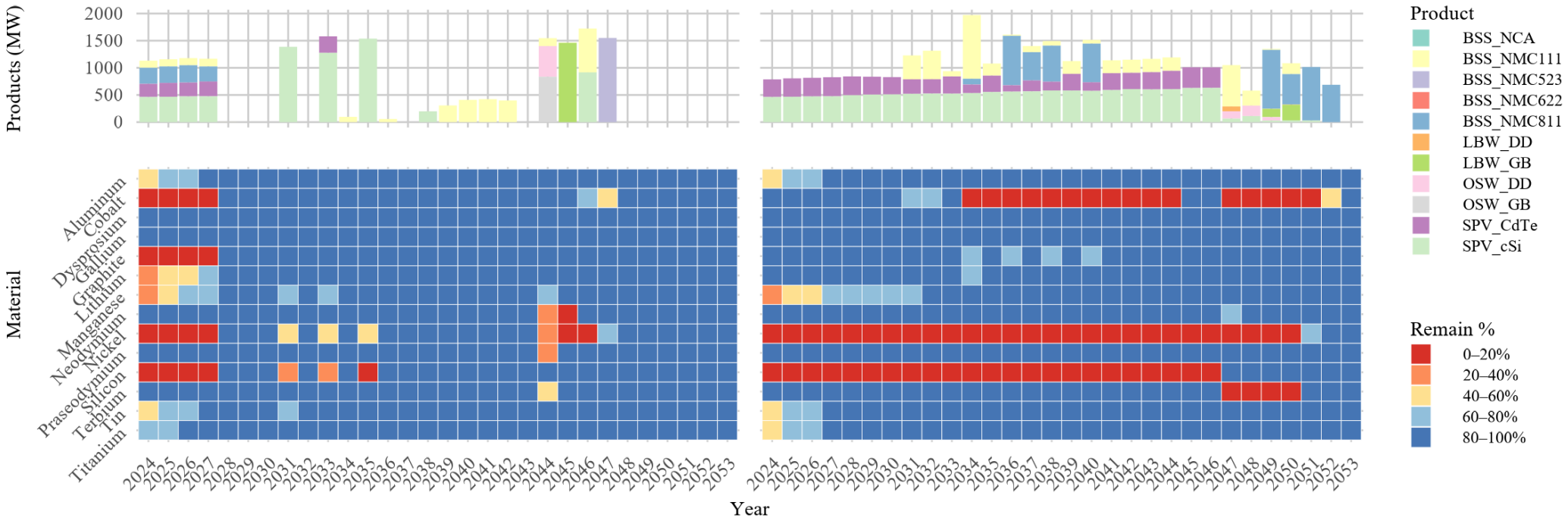


Fig. 3. Planned product by technology (top) and remaining material availability (bottom) for baseline Low and High scenarios.

- Low baseline:
  - Pre-2031: SPV/BSS constrained by silicon & nickel bottlenecks.
  - 2040s: shift to LBW & OSW as costs fall and materials accumulate.
  - Mix of gearbox & direct-drive OSW to balance nickel vs. neodymium use.
- High baseline:
  - Continuous load growth → sustained SPV build to 2046 (fast build, low material use).
  - Storage shifts from NMC 111 (cobalt) to NMC 811 (nickel).
  - 2047: SPV insufficient → add LBW & OSW (direct-drive favored for nickel limits).

Results

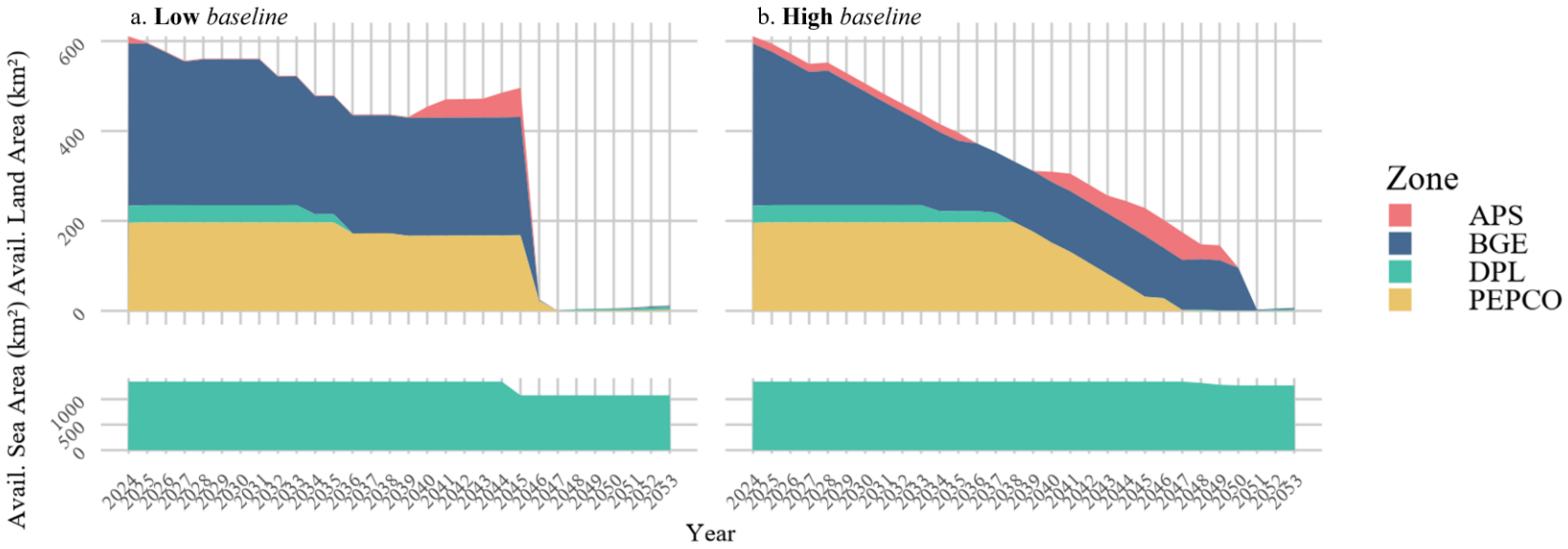


Fig. 4. Field availability at the beginning of each year: land (top) and offshore (bottom) area for new deployment.

- Post-2049: land outside BGE fully allocated → spatial constraint binds
- Response: shift to BSS (low land need, short lead time)
- Materials: move toward NMC 811 (less cobalt) while keeping some NMC 111 for balance

Scenario Comparisons

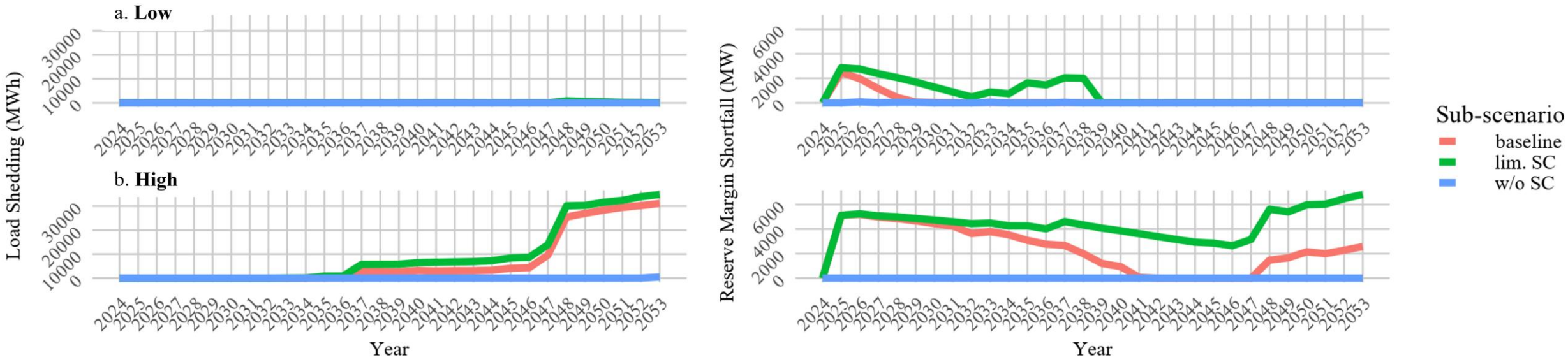


Fig. 5. Yearly load shedding and reserve margin shortfalls across all scenarios.

- **w/o SC (relaxed):**
  - Unlimited materials & no lead times → just-in-time planning.
  - Low: LBW added in 2048 (cheap & land available).
  - High: sustained load growth → more OSW after 2045.
- **lim. SC (constrained):**
  - Critical material limits (esp. rare earths) → no new wind after 2045.
  - SPV (c-Si) favored due to steady silicon supply.
  - BSS shift: NMC 811 → NMC 111 (less nickel).
  - Stronger shift in High case due to higher load stress.

- **Reliability:**
  - w/o SC: almost no issues (flexible, fast build)
  - baseline: early shortfalls (2025 retirements), mostly resolved by 2041
  - lim. SC: persistent reserve margin violations; load shedding in High from 2037 onward; major gap after 2048 NGCC retirements remains unresolved

Key Takeaways

- Ignoring supply chains underestimates costs and reliability risks.
- Low vs. High: modest cost increase in Low; severe shortfalls and load shedding in High.
- Just-in-time planning is infeasible — requires earlier, strategic investments and possible lifetime extensions.
- Constraints can overturn cost advantages — diversification across technologies is critical.
- Anticipatory planning needed to avoid over-correction and material exhaustion.